

HACKATHON INNOVATION DECK

A.I. RURAL FINANCE

Empowering Rural Merchants & Farmers with Alternative Credit Decisioning

MEET THE TEAM			
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Before this Idea: The Credit Gap

The harsh reality of agricultural micro-finance in rural areas

Traditional Banking Hurdles

1. Collateral Vacuum: Farmers do not have land titles or capital assets

to pledge, immediately disqualifying them from bank micro-loans.

2. No Paper Trail: Local trade is cash-based. Without bank statements

or tax returns, creditworthiness evaluation is impossible.

3. Geographical Isolation: Rural branches are sparse, making in-person

audits expensive and slow for microfinance institutions.

Social & Financial Consequence

▣ Debt Traps: Farmers resort to local informal money lenders (loan sharks)

who charge compound interest rates up to 36% to 60% per annum.

▣ Delayed Relief: When regional monsoon failures occur, wait-times

for government agricultural reliefs take months, destroying crops.

▣ Low Digital Trust: Language complex jargon in scheme paperwork

leads to low conversion and high rejection rates.

After this Solution: Empowering Benefits

Transforming credit scoring and access through humanized tech

Direct Benefits to Rural Borrowers

▣ Financial Footprint: Instantly builds a digital credit trail using standard

UPI QR payments, broadcast live to bank decision portals.

▣ Credit Quiz Boosts: User earns a +5% Credit Borrowing Limit increase

by completing interactive financial literacy games directly in the app.

▣ Immediate Drought Overdraft: Instantly claim Rs. 10,000 pre-approved,

zero-interest emergency monsoon funds when weather delays threaten crops.

Benefits to Lending Institutions

▣ Lower Default Risks: AI predictive classifiers evaluate monthly cash margins

and business trends continuously, reducing defaults by 25%.

▣ SHG Trust Multipliers: Self Help Group peer rankings trigger limit boosts

backed by collective accountability, minimizing high-risk bad debts.

▣ Direct KCC Subsidy: Integrated Kisan Credit Card subsidy logic slashes

micro-loan interest from 8.5% to 4.0%, backed by state reimbursement.

The Complete Tech Stack

Robust integration of modern frameworks and machine learning

1. Frontend Architecture

- Next.js 15 (TypeScript)
- TailwindCSSCustom Styling
- HTML5 Web Speech Synthesis API
- Interactive Score Simulators
- Responsive Tour Guide Player
- Local Cache Sync Badges

2. Backend API & Database

- Node.js & Express API Server
- WebSockets for Live Payments
- MongoDB & Mongoose Schemas
- Redis Cache Sync Layer
- Secure Token JWT Auth
- Loan Application Pipeline

3. Machine Learning Engine

- Python & FastAPI (Uvicorn)
- Scikit-Learn Decision Models
- Mandi price trend forecasts
- Pandas Data Frame processing
- Synthetic Credit Risk Training
- Alternative validation metrics

10 Theme-Specific Features

Complete list of upgrades built for the prototype

1. UPI Audio Soundbox: Speech payment alerts broadcast live.

2. Mandi Price Tracker: Wholesale rate cards for main crops.

3. Mandi Trend Forecaster: 30-day directional crop projections.

4. SHG Peer Scores: +15% borrowing limit community boosts.

5. Interactive Literacy Quizzes: rewards +5% credit eligibility.

6. Warehouse Receipt Collateral: +Rs 2,500 borrowing cap per bag.

7. Monsoon Overdraft: Instant, zero-interest Rs. 10,000 emergency fund.

8. Kisan Credit Card Gateway: Slashes interest from 8.5% to 4.0%.

9. Offline Sync Indicators: Pulsing indicators for cache resilience.

10. Leasing ROI Calculator: Daily tractor lease ROI savings calculator.

Social & Commercial Impact

Delivering measurable financial inclusion outcomes

Social Inclusion & Empowerment

▣ Women-led SHG Promotion: Group guarantees encourage

marginalized women-led cooperatives to borrow safely.

▣ Vernacular Equality: Sound-driven alerts bypass text literacy

hurdles, empowering older, non-tech-savvy farmers.

▣ Climate Resilience: Emergency weather overdraft prevents crop

failures from translating into severe structural debts.

Commercial Feasibility & Security

▣ Lower Non-Performing Assets (NPA): Scikit-Learn risk assessment

checks cash flow consistency and deters over-borrowing.

▣ Offline Resilience: Local indexing sync badge prevents transaction

failures in regions with weak mobile internet coverage.

▣ Transparent Auditing: Secure digital ledger allows auditing of KCC

interest subsidies and warehouse receipt validations.